

Qns	Answer	Marks
5ai	Binding energy of Mo = $95 \times 8.09 = 768.55 \text{ MeV}$ Binding energy of La = $139 \times 7.92 = 1100.88 \text{ MeV}$ Energy released = Binding energy of Mo + Binding energy of La – Binding energy of U Therefore $182 = 768.55 + 1100.88 - \text{Binding energy of U}$ Binding energy of U = $1687.43 \text{ MeV} = 1690 \text{ MeV}$	C1 (either one) M1 A1
5aii	Binding energy = $\Delta m c^2$ $\Delta m = \frac{1687.43 \times 10^6 \times 1.6 \times 10^{-19}}{(3.0 \times 10^8)^2}$ = $3.00 \times 10^{-27} \text{ kg}$	C1 A1
5bi	${}^{42}_{19}\text{K} \rightarrow {}^{42}_{20}\text{Ca} + {}^0_{-1}\text{e} + \bar{\nu}$ Also accept: “neutrino”, “anti-neutrino” and ν	A1
5bii	From Fig. 5.2, time taken for number of nuclei to drop from N_0 to $0.25 N_0$ is 25 hours. $2t_{1/2} = 25$ $t_{1/2} = 12.5 \text{ hours}$	A1
5biii	Graph of Ca is a reflection of the graph for K in the $N = 0.5 N_0$ line. Graphs will intersect at $N = 0.5 N_0$	P1
5biv	When $N(\text{Ca}) = 4 N(\text{K})$, $N(\text{K}) = 0.2 N_0$ From the graph, this occurs when $t = 29 \text{ hours}$	M1 A1

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c(e)	acceleration proportional to displacement (from a fixed point)	D1